

Remark 2.14. In $PL_{[0,1]}$ s, the evaluation of formulas with connective $\&$ is traditionally extended as $\|a\&b\| = T(\|a\|, \|b\|) = \|a\| * \|b\|$ and the evaluation of formulas with connective $\wedge$ is extended as $\|a \wedge b\| = \min\{\|a\|, \|b\|\}$. For keeping the unification of form of the normal encoding function, we extend the evaluation of formulas with connective $\wedge$ as $\|a \wedge b\| = T(\|a\|, \|b\|) = \|a\| * \|b\|$, which will not cause any confusion in this paper.

2.3. Model Checking

Besnard and Doutre [12] give a method of model checking. The method aims to translate an AF as a propositional formula in the PL_2 , such that the models of the propositional formula are corresponding to extensions of the AF under a kind of semantics. In fact, the core of each model is an extension of the AF and vice versa. The translating functions depend on various semantics. The mainly results in the chapter “model checking” in [12] are listed as the following, where each Φ is a translating function.

Let $AF = (A, R)$ be an argumentation framework.

1. Translating the AF to the PL_2 for conflict-free semantics.

A set $S \subseteq A$ is conflict-free iff S is the core of a model of any formula below:

$$\bigwedge_{a \in A} (a \rightarrow \bigwedge_{b: (b,a) \in R} \neg b); \quad (2.1)$$

$$\bigwedge_{a \in A} (a \rightarrow \bigwedge_{b: (a,b) \in R} \neg b); \quad (2.2)$$

$$\bigwedge_{(a,b) \in R} (\neg a \vee \neg b). \quad (2.3)$$

2. Translating the AF to the PL_2 for stable semantics.

A set $S \subseteq A$ is a stable extension iff S is the core of a model of the formula below:

$$\bigwedge_{a \in A} (a \leftrightarrow \bigwedge_{b: (b,a) \in R} \neg b). \quad (2.4)$$

This formula is come up with by Creignou in [29] to characterize the kernels of a graph, and is applied in [12] to characterize the stable extensions of an AF .

3. Translating the AF to the PL_2 for admissible semantics.

A set $S \subseteq A$ is a admissible set iff S is the core of a model of the formula below:

$$\bigwedge_{a \in A} ((a \rightarrow \bigwedge_{b: (b,a) \in R} \neg b) \wedge (a \rightarrow \bigwedge_{b: (b,a) \in R} (\bigvee_{c: (c,b) \in R} c))). \quad (2.5)$$

4. Translating the AF to the PL_2 for complete semantics.

A set $S \subseteq A$ is a complete extension iff S is the core of a model of the formula below:

$$\bigwedge_{a \in A} ((a \rightarrow \bigwedge_{b: (b,a) \in R} \neg b) \wedge (a \leftrightarrow \bigwedge_{b: (b,a) \in R} (\bigvee_{c: (c,b) \in R} c))). \quad (2.6)$$